CAMERON VOLOSHIN

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RL Research & ML Engineering · Shipping Real-Time ML on Edge · Ph.D., Caltech

SUMMARY

Reinforcement-learning researcher and engineer with a Caltech PhD and publications in policy optimization, safe/constrained RL, and off-policy evaluation. Designed and shipped the first learned driving model to run on-vehicle for Ford's self-driving fleet — a milestone for the program. Architects the full ML stack behind it — from multi-cluster training built for both production scale and fast research iteration, to optimized, low-latency CPU-bounded inference.

SELECTED EXPERIENCE

Latitude AI (Ford — Self-Driving) — *Staff Research Scientist / Engineer, Motion Planning* Jun 2023 – Present

Remote

- **Lead ML for Ford's self-driving motion planner** — a 4+ engineer team owning the full lifecycle from data generation to resource-constrained on-vehicle deployment. Conceived and shipped the first learned forecasting system (replacing the heuristic stack) and the first ML planner to run on-vehicle, which drives the car over 30% of the time.
- **Rebuilt** the distributed ML training pipeline, enabling multi-cluster GPU training and parallelized data loading; further profiled and removed CPU/GPU bottlenecks (custom operators) for >2× total training throughput — pragmatically balancing GPU-training speed against CPU-inference constraints.

Activision — *Machine Learning Consultant* Jan 2021 – Jan 2022

Los Angeles, CA

- **Oversaw** recommendation systems for message/ad personalization (contextual bandits, collaborative filtering) and off-policy-evaluation / A-B backtesting tools; led hiring of a data-science team and mentored interns.

EsportsOne / CivicScape — *Machine Learning Engineer / Researcher* 2017 – 2020

LA / Chicago

- **Built** real-time computer-vision and deep-learning pipelines at scale (CNNs, GANs, OCR; OpenCV/TensorFlow) on AWS (EC2/Lambda/SNS, Docker), plus spatio-temporal crime prediction with distributed EC2 preprocessing and PostGIS.

SELECTED PUBLICATIONS

- **Voloshin, C.**, Verma, A., Yue, Y. "Eventual Discounting Temporal Logic Counterfactual Experience Replay." ICML 2023.
- **Voloshin, C.**, Le, H. M., Chaudhuri, S., Yue, Y. "Policy Optimization with Linear Temporal Logic Constraints." NeurIPS 2022.
- **Voloshin, C.**, Jiang, N., Yue, Y. "Minimax Model Learning." AISTATS 2021.
- **Voloshin, C.**, Le, H. M., Jiang, N., Yue, Y. "Empirical Study of Off-Policy Policy Evaluation for Reinforcement Learning" (COBS benchmark). NeurIPS 2021 Datasets & Benchmarks.
- **Le, H. M., Voloshin, C.**, Yue, Y. "Batch Policy Learning under Constraints" (FQE). ICML 2019.
- **Shah, A., Voloshin, C.**, et al. "Deep Policy Optimization with Temporal Logic Constraints" (CyclER). arXiv 2024.

TECHNICAL SKILLS

Reinforcement Learning: policy optimization, safe/constrained RL, offline/batch RL, off-policy evaluation (COBS, FQE), LTL-constrained RL, training environments & evaluations, PAC analysis

RL Infrastructure & ML Eng: Ray distributed training / experiment management on GPU clusters, PyTorch, torch compilation, async/concurrent Python, GPU profiling & benchmarking, ONNX + custom C++ operators, low-latency inference

Systems & Tooling: Python, C++ (production ML/control), distributed systems, typed API design, testing & code quality; Dagster, W&B, BigQuery, AWS, Docker, PostGIS, Protobuf

Domains: safety-critical real-time ML, autonomous vehicles / motion planning, recommendation systems, computer vision

EDUCATION & HONORS

Ph.D., Computational & Mathematical Sciences — California Institute of Technology Jun 2023

Advisor: Prof. Yisong Yue. Thesis: "Guaranteed Policy Performance in Reinforcement Learning."

B.S., Applied & Computational Mathematics — California Institute of Technology Jun 2017

Fellowships & Awards: NSF Graduate Research Fellow (2022–2026); Caltech Kortschak Fellow (2020–2022); Caltech Frederick J. Zeigler Memorial Award (2015).